

RING OSCILLATOR

1. Introduction

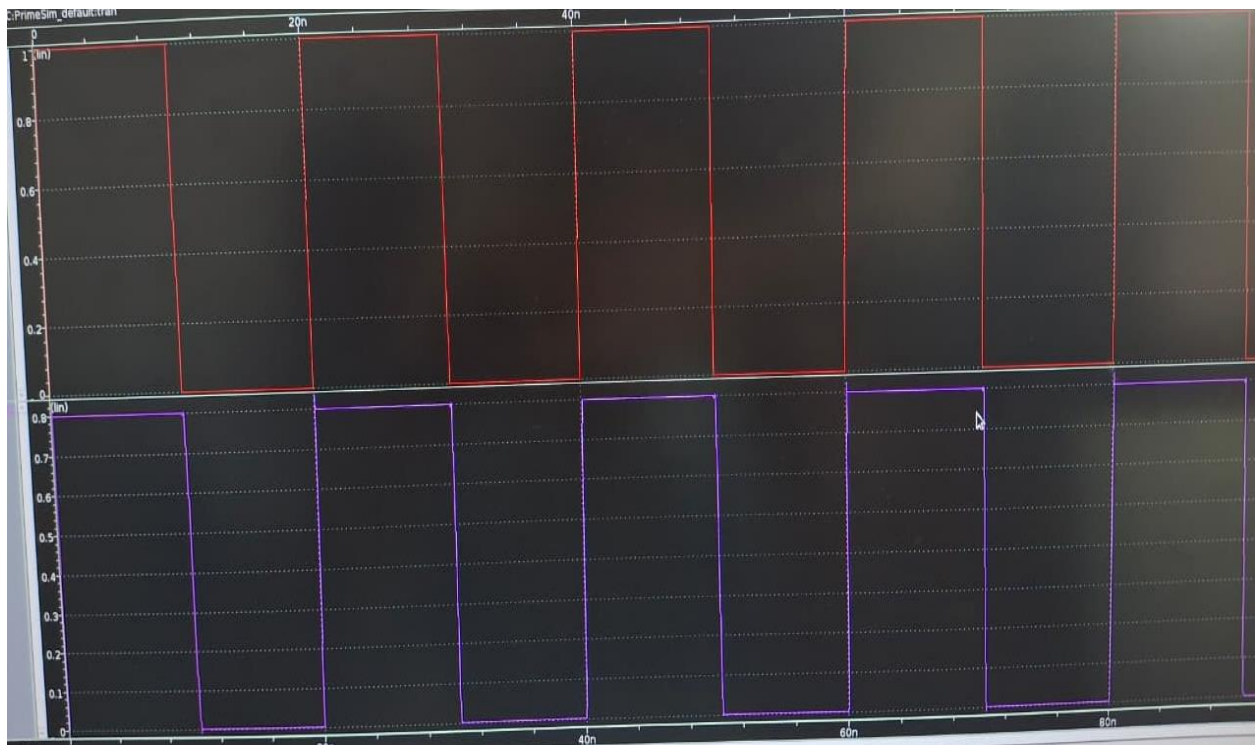
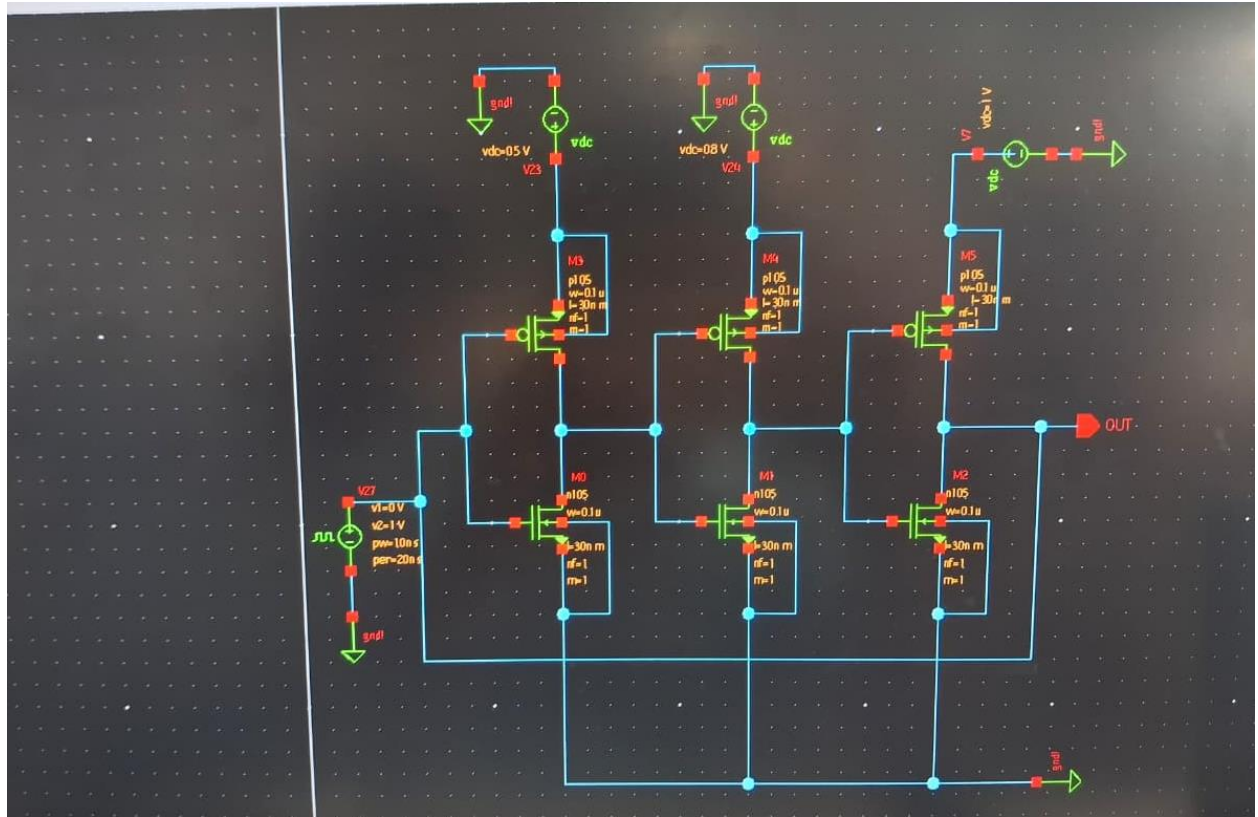
- A ring oscillator is a fundamental electronic circuit composed of an odd number of inverting stages (e.g., CMOS inverters) connected in a closed loop. Due to the odd number of inversions and their propagation delay, the output signal oscillates perpetually between logic HIGH and LOW. Ring oscillators are widely used for on-chip clock generation, process monitoring, and as timing references in VLSI and mixed-signal systems.

2. Circuit Diagram

- The typical ring oscillator consists of 3, 5, or more inverter stages connected in series, with the output of the last inverter fed back to the first input.

3. Principle of Operation

- Upon powering up, noise or initial logic states trigger a transition in one inverter. This change propagates sequentially through the inverter chain, each introducing a time delay. Because the number of inverters is odd, the loop inverts its input, creating perpetual oscillations.
- The oscillation frequency is determined by:
$$F = 1 / 2NTP$$
 - N – Number of stages
 - TP – Propagation delay per inverter.



4. Importance of Odd Inverters & Feedback Roles

- **An odd number of inverters is essential:** it guarantees negative feedback. The output always inverts the input, never settling into a stable logic state.
- **If even stages are used:** output equals input after propagation—causing the loop to stabilize (no oscillation); this is positive feedback and can act like a latch, not an oscillator.
- **Negative feedback (odd stages):** forces the loop to continually change state, causing the desired oscillation.
- **Positive feedback (even stages):** leads to bistable (latch-like) operation, preventing oscillation.

5. Role in Capacitance-to-Digital Converter

- **Reference Clock Source:** The ring oscillator serves as the timing engine for CDC circuits, enabling precise sampling and digital conversion of analog capacitance changes
- **Capacitance Sensing:** By integrating a variable capacitance within the oscillator's load (at one or several inverter nodes), its propagation delay and therefore oscillation frequency are modulated. Counting oscillation cycles allows direct digital conversion of capacitance.
- **PVT Monitoring:** Frequency variations due to supply voltage, process, and temperature enable on-chip calibration and compensation, improving CDC accuracy.

6. Parameter Selection, EDA Setup & Supply Voltage Effects:

- **Key Parameters**

- Number of Stages: 3, 5, or more—the higher the count, the lower the frequency due to cumulative delay.
- Supply Voltage VDD: Sets maximum output swing and affects inverter speed. Typical values are 1.2V, 1.8V, or 3.3V.
- Load Capacitance: Directly influences delay, hence the oscillation frequency.

- **EDA Simulation Setup**

- VDD: Connect a DC source to all inverter rails.
- Input/Trigger: Not required; oscillation starts automatically due to inherent noise and feedback.
- Transient Analysis: Probe at any inverter output node to observe the generated square waveform.

- **Voltage and Stage Effects**

- Higher VDD: Improves drive strength, increases frequency, and enhances noise margin.
- Lower VDD: Slows inverter response, decreases frequency, and below threshold, can halt oscillation.
- Increasing stages: Adds extra delay, lowers output frequency

7. Effects of Even and Odd Stages

Parameter	Behavior
Odd stages	Negative feedback, forced oscillation
Even stages	Positive feedback, static latch, no oscillation
More stages (n)	Lower frequency (higher delay per cycle)
Higher VDD	Higher frequency, better margins
Lower VDD	Lower frequency, possible instability

8. Applications

- On-chip Clock Generation
- Process, Voltage, and Temperature Sensors
- Capacitance Measurement and CDC Integration
- Built-In Self Test BIST
- Jitter/Timing Analysis

9.Waveform and Measurement

- The output is a periodic square waveform with frequency determined by the design parameters.
- Measure period and frequency at an inverter output node during simulation